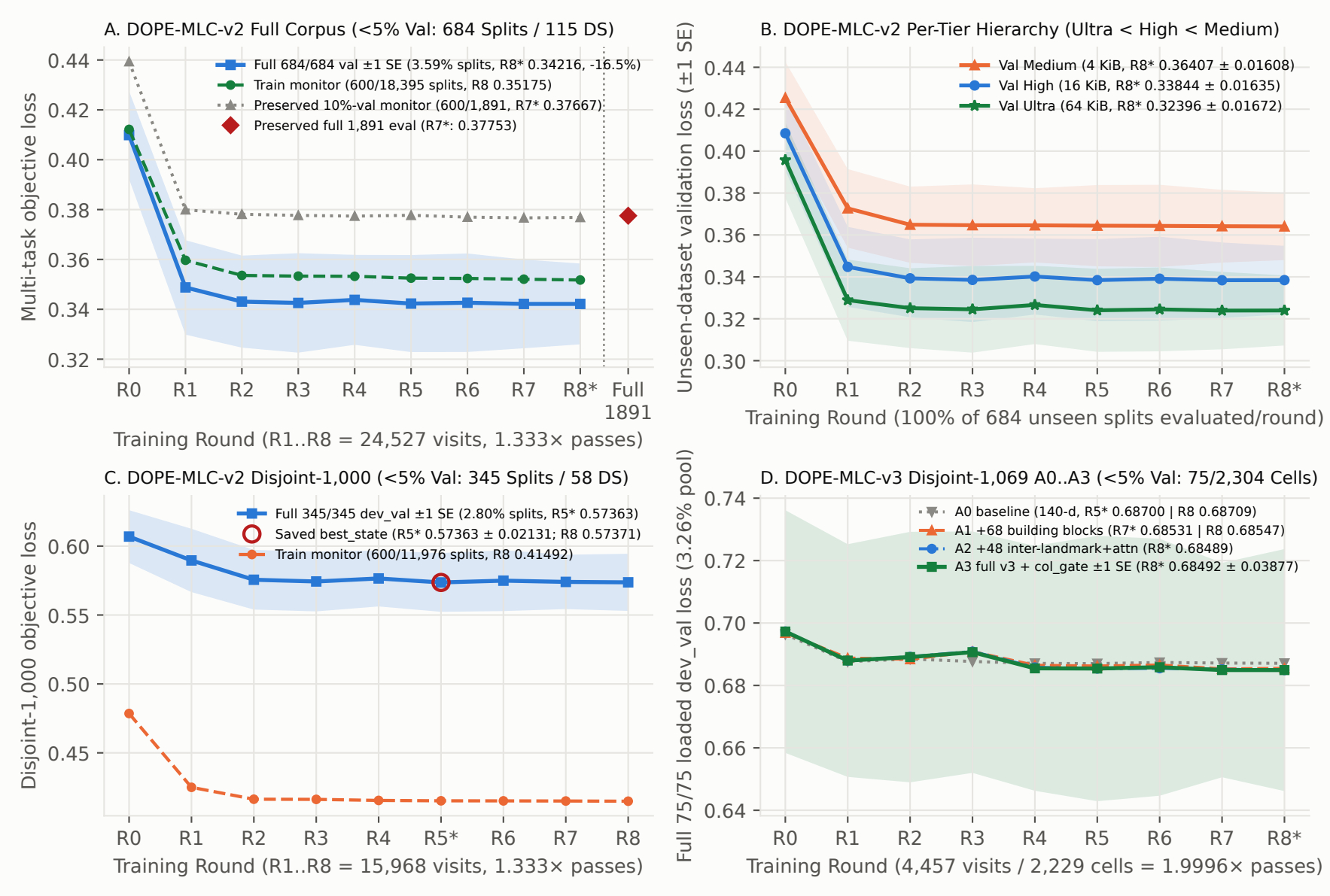
A. DOPE-MLC-v2 Full Corpus (<5% Val: 684 Splits / 115 DS)
Full 684/684 val ±1 SE (3.59% splits, R8* 0.34216, -16.5%)
Train monitor (600/18,395 splits, R8 0.35175)
Preserved 10%-val monitor (600/1,891, R7* 0.37667)
Preserved full 1,891 eval (R7*: 0.37753)
Multi-task objective loss
0.44
0.42
0.40
0.38
0.36
0.34
0.32
R0
R1
R2
R3
R4
R5
R6
R7
R8*
Full 1891
Training Round (R1..R8 = 24,527 visits, 1.333× passes)
B. DOPE-MLC-v2 Per-Tier Hierarchy (Ultra < High < Medium)
Val Medium (4 KiB, R8* 0.36407 ± 0.01608)
Val High (16 KiB, R8* 0.33844 ± 0.01635)
Val Ultra (64 KiB, R8* 0.32396 ± 0.01672)
Unseen-dataset validation loss (±1 SE)
0.44
0.42
0.40
0.38
0.36
0.34
0.32
0.30
R0
R1
R2
R3
R4
R5
R6
R7
R8*
Training Round (100% of 684 unseen splits evaluated/round)
C. DOPE-MLC-v2 Disjoint-1,000 (<5% Val: 345 Splits / 58 DS)
Full 345/345 dev_val ±1 SE (2.80% splits, R5* 0.57363)
Saved best_state (R5* 0.57363 ± 0.02131; R8 0.57371)
Train monitor (600/11,976 splits, R8 0.41492)
Disjoint-1,000 objective loss
0.60
0.55
0.50
0.45
R0
R1
R2
R3
R4
R5*
R6
R7
R8
Training Round (R1..R8 = 15,968 visits, 1.333× passes)
D. DOPE-MLC-v3 Disjoint-1,069 A0..A3 (<5% Val: 75/2,304 Cells)
A0 baseline (140-d, R5* 0.68700 | R8 0.68709)
A1 +68 building blocks (R7* 0.68531 | R8 0.68547)
A2 +48 inter-landmark+attn (R8* 0.68489)
A3 full v3 + col_gate ±1 SE (R8* 0.68492 ± 0.03877)
Full 75/75 loaded dev_val loss (3.26% pool)
0.74
0.72
0.70
0.68
0.66
0.64
R0
R1
R2
R3
R4
R5
R6
R7
R8*
Training Round (4,457 visits / 2,229 cells = 1.9996× passes)